Bandits and Gaussian Processes

MSEN 489/655 - Materials Design Studio

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Outline

1. Introduction
2. Exercise
3. Bandits
4. Gaussian Processes

Introduction

Lab Objectives

- Why "intelligent experiments" matter in materials?
- Exercise: full factorial vs Bayesian optimization (BO)
- Bandits: rock-paper-scissors game
- Gaussian processes (GPs): hands-on exercises

Motivation: Experiments are Expensive

- Materials design is often a **black-box mapping**:

$$\text{composition} + \text{processing} \rightarrow \text{microstructure} \rightarrow \text{properties}$$

- Each evaluation can require:
  - synthesis + processing (hours–days),
  - characterization (SEM/XRD/DSC, etc.),
  - post-processing + analysis.
- So: **every run matters**. We want to learn the most per experiment.

Common objectives:

- **Optimization**: find best-performing material
- **Understanding**: map the response surface $f(x)$

Common constraints:

- Feasibility (phase stability, safety)
- Multiple objectives (strength vs ductility)

Active Learning Loop

Given design space, objectives & constraints, do this three iteratively:

- **Design:** Choose next experiments
- **Execute:** Synthesize & measure
- **Model:** Update surrogate & uncertainty

From today's topic: "Design" is where **bandits** and **bayesian optimization** live, and **GPs** are a common modeling choice.

For your Projects

1. Define:
   - Design Space (e.g., composition fractions, temperature, time, additive)
   - Objectives (e.g., conductivity, hardness, cycle life)
   - Constraints (feasible region, safety, budget)
2. Find data source.
3. Active learning loop.

Exercise

Problem Setup: Expensive Black-box Objective

We will model a "materials score" function

$$y = f(x) + \varepsilon, \qquad x \in [0,1]^2$$

- x might be: *composition fraction* (x_1) and *processing temperature* (x_2)
- f is **unknown**, nonlinear, and costly to query
- ε represents measurement noise

Two strategies for experiment selection

- **Random**: choose what point you want to experiment
- **BO-guided**: sequentially choose points using a surrogate + uncertainty

Activity: Random vs BO on a Toy "Materials Score"

Instructions

- Select oracle $f(x_1, x_2)$ and objective (minimization or maximization).
- Initial data: compute $f(0,0), f(0,1), f(1,0), f(1,1)$.
- Budget: **5** oracle queries total.
- For each strategy, output optimal point, regret per iteration and a quick sketch of the surface.

Bandits

What is a bandit?

Definition

A bandit is a sequential decision problem where you:

1. choose an action/arm/**experiment** a_t
2. observe only the reward/**measurement** of the chosen arm r_t
3. update your strategy to maximize total reward

Key tension

Exploration vs exploitation:

- exploit what looks best now
- explore to learn what might be better

Regret: Standard Metric for Learning

Stochastic bandit (one common setting)

Each arm a has unknown mean reward μ_a. The best arm has $\mu^* = \max_a \mu_a$.

Cumulative regret after T rounds

$$R_T = T\mu^* - \sum_{t=1}^{T} r_t$$

Low regret means: you behave almost as well as always playing the best arm.

Why regret matters in lab workflows

If you are running experiments *while also delivering results*, you want to avoid wasting too many trials on bad settings.

Activity: Rock-paper-scissors as game theory

Play 10 rounds against random computer

Keep tally:

- Arms: a_1 is rock, a_2 is paper, a_3 is scissors
- Rewards: 1 if win, 0 it tie, -1 if loose. $\mu^* = 1$
- Calculate regret per iteration

Play 10 rounds against a peer on your side

Keep tally:

- Arms: a_1 is rock, a_2 is paper, a_3 is scissors
- Rewards: 1 if win, 0 it tie, -1 if loose. $\mu^* = 1$
- Calculate regret per iteration

Gaussian Processes

Why Gaussian processes in experiment selection?

GPs give you two things you really want

- a prediction (mean) for $f(x)$
- an uncertainty estimate (variance) for $f(x)$

Acquisition functions use both to decide where to measure next.

A GP is a distribution over functions

$$f(\cdot) \sim \mathcal{GP}(m(\cdot), k(\cdot, \cdot))$$

- $m(x)$: prior mean
- $k(x, x')$: kernel/covariance

Kernels

Common kernels (examples)

- Squared exponential / RBF: $k(x, x') = \sigma_f^2 \exp(-\frac{\|x-x'\|^2}{2\ell^2})$
- Matérn: less smooth, often more realistic
- Periodic: repeating patterns
- Linear / polynomial: global trends

GP regression: posterior mean and uncertainty

Given data $X = [x_1, \ldots, x_n]$ and y

With Gaussian noise σ_n^2, define K where $K_{ij} = k(x_i, x_j)$.

Prediction at a new point x_*

Let $k_* = [k(x_*, x_1), \ldots, k(x_*, x_n)]^\top$. Then:

$$\mu(x_*) = k_*^\top (K + \sigma_n^2 I)^{-1} y$$

$$\sigma^2(x_*) = k(x_*, x_*) - k_*^\top (K + \sigma_n^2 I)^{-1} k_*$$

- mean: best guess
- variance: how unsure we are

Activity

- Play with GP and Kernels on Python
- Calculate some by hand

Thank You!

Questions?